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Review article

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The clinical landscape of HIF2α inhibitors in oncology

Eddy Saad[1,2,3,5], Marc Machaalani[1,3,5], David F. McDermott[1,3,4] & Toni K. Choueiri[1,3]

Abstract

Hypoxia-inducible factor 2α (HIF2α; also known as endothelial PAS domain-containing protein 1) had long been considered an undruggable transcription factor until the discovery of an allosteric pocket within its PAS-B domain enabled the development of selective small-molecule antagonists. Belzutifan, the first-in-class HIF2α inhibitor, has since demonstrated substantial efficacy in patients with von Hippel–Lindau (VHL) disease-associated tumours, sporadic clear-cell renal cell carcinoma (ccRCC), and pheochromocytoma or paraganglioma, thereby validating HIF2α as a therapeutic target in patients with cancer. In this Review, we summarize the biology of the VHL–HIF signalling pathway, the structural basis for HIF2α druggability and the clinical milestones leading to the multiple regulatory approvals of belzutifan. We also highlight emerging data on other small-molecule inhibitors, RNA interference approaches and indirect modulators that have the potential to expand the scope of HIF pathway suppression. Combination strategies offer opportunities to enhance efficacy and overcome resistance; however, key challenges remain, including the identification of predictive biomarkers, mechanisms of primary and acquired resistance, and optimal management approaches for on-target toxicities such as anaemia and hypoxia. Finally, we discuss the increasing potential of HIF2α inhibition beyond kidney cancer, including its role in hypoxia-adapted malignancies, and outline priorities for future translational and clinical research.

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[1]Department of Medical Oncology, Dana–Farber Cancer Institute, Boston, MA, USA. [2]Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA, USA. [3]Harvard Medical School, Boston, MA, USA. [4]Division of Medical Oncology, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA, USA. [5]These authors contributed equally: Eddy Saad, Marc Machaalani. e-mail: toni_choueiri@dfci.harvard.edu

Key points

- Identification of an allosteric pocket in the Per-ARNT-Sim (PAS)-B domain of hypoxia-inducible factor 2α (HIF2α) rendered this transcription factor druggable, enabling selective small-molecule inhibition of a previously intractable oncogenic driver.
- Belzutifan, the first-in-class allosteric HIF2α inhibitor, has demonstrated durable clinical activity in von Hippel–Lindau (VHL) disease-associated tumours and sporadic clear-cell renal cell carcinoma (ccRCC), establishing HIF2α as a validated therapeutic target.
- HIF2α inhibition produces predictable mechanism-related, on-target toxicities, particularly anaemia and hypoxia, requiring proactive monitoring and dose-adapted management.
- Primary and acquired resistance to HIF2α inhibitors reflect variable degrees of tumour dependence on HIF2 signalling and mutations affecting drug–target engagement, highlighting the need for predictive biomarkers.
- Combination strategies with VEGF-targeted therapy, immune-checkpoint inhibitors and modulators of other pathways are showing promise for enhancing the efficacy of and overcoming resistance to HIF2α inhibitors in ccRCC.
- Emerging data support broader applications of HIF2α inhibition beyond ccRCC, warranting continued translational and clinical investigations.

Introduction

The hypoxia-inducible factor (HIF) signalling pathway has been implicated in tumorigenesis, especially in the context of clear-cell renal cell carcinoma (ccRCC)[1]. The development of small-molecule inhibitors of HIF2α — a target long perceived to be 'undruggable' — has gathered remarkable momentum over the past decade, culminating in regulatory approvals of belzutifan for the treatment of von Hippel–Lindau (VHL) syndrome-associated tumours[2], advanced-stage sporadic ccRCC[3] and, more recently, pheochromocytoma and paraganglioma (PPGL)[4]. These breakthroughs have sparked interest in refining the clinical use of HIF2α inhibitors, including designing novel agents, identifying biomarkers of response and managing on-target toxicities. Moreover, emerging evidence indicates that other genetic alterations affecting the VHL–HIF pathway and the tricarboxylic acid (TCA) cycle might sensitize a broader range of tumour types to this therapeutic strategy, opening the door to more widespread applications in oncology[5].

Herein, we first review the fundamental biology of the hypoxia-sensing pathway and research efforts that led to the development of small-molecule HIF2α inhibitors. We then summarize major clinical milestones underpinning the use of these agents for the treatment of ccRCC, emphasizing key trials and the regulatory approvals granted to date. Next, we assess current strategies to improve the efficacy of HIF2α inhibitors through combination with other therapies and the development of novel HIF2α antagonists, and discuss emerging data on biomarkers of response, toxicities and resistance. Finally, we examine the growing potential of these agents beyond ccRCC, including other tumour types and possible non-cancer indications.

The oxygen-sensing VHL–HIF pathway

Metazoans have evolved mechanisms to sense and adapt to changes in oxygen availability, with these processes being governed by the VHL–HIF pathway[6]. The central actor in this pathway is HIF, a heterodimeric transcription factor composed of a constitutively expressed β-subunit (HIF1β; also known as aryl hydrocarbon receptor nuclear translocator (ARNT)) and an oxygen-regulated α-subunit (either HIF1α or HIF2α; the latter also known as endothelial PAS domain-containing protein 1 (EPAS1))[7]. Both the α-subunit and β-subunit share a similar structure containing a basic helix–loop–helix (bHLH) region at their N terminus that enables DNA binding at hypoxia-response elements (HREs), a C-terminal transactivation domain (C-TAD) regulating gene expression, as well as two intervening Per-ARNT-Sim (PAS) domains — PAS-A and PAS-B — that interact with their heterodimeric counterparts to support dimerization[8]. The HIFα subunit additionally possesses an oxygen-dependent degradation (ODD) domain that can be hydroxylated, in the presence of oxygen, by prolyl hydroxylase domain-containing (PHD) proteins, also called Egl nine homologues (EGLNs)[9,10]. This hydroxylation event creates a recognition motif for the protein product of the *VHL* gene, pVHL (also known as VHL disease tumour suppressor), which functions as the substrate-receptor component of an E3 ubiquitin ligase complex[11,12]. Once bound to pVHL, the hydroxylated HIFα subunit is ubiquitinated and subsequently directed to the proteasome for degradation, thereby preventing accumulation of HIF transcription factors when oxygen levels are sufficient[11,12] (Fig. 1).

By contrast, under hypoxic conditions, the activity of oxygen-dependent PHD/EGLN enzymes is reduced and, thus, the hydroxylation of HIFα subunits is markedly suppressed, which enables them to evade degradation and accumulate in the cytoplasm[9,10]. The stabilized HIFα subunits can then translocate to the nucleus, where they dimerize with HIF1β and bind to HREs, initiating the transcription of multiple target genes[13]. These genes encode proteins involved in angiogenesis (such as vascular endothelial growth factor (VEGF))[14], erythropoiesis (including erythropoietin (EPO))[15], glucose metabolism (for example, glucose transporter type 1 (GLUT1) and various glycolytic enzymes)[16,17] as well as pathways governing cell survival and proliferation (such as cyclin D1)[18–20] (Fig. 1).

However, these same pathways can also be activated at non-hypoxic intracellular oxygen levels when the VHL–HIF pathway is pathologically dysregulated, a phenomenon known as a 'pseudo-hypoxia'[21]. A quintessential example of this effect is provided by loss of *VHL* function[6]: in its inherited form, germline *VHL* inactivation causes VHL syndrome, a condition characterized by the development of various angiogenic tumours, including central nervous system (CNS) haemangioblastomas, pheochromocytomas and ccRCCs, as well as pancreatic neuroendocrine tumours (pNETs)[22]. Somatic *VHL* mutations and/or epigenetic silencing of this gene owing to aberrant promoter methylation are also frequently observed in the much more common sporadic ccRCCs[23,24]. In both scenarios, the absence of functional pVHL prevents degradation of HIFα subunits, leading to constitutive activation of HIF even under normoxic conditions[25,26]. This inappropriate HIF accumulation and signalling recapitulates many of the cellular changes associated with true hypoxia, thereby promoting angiogenesis and activating other oncogenic signalling pathways[27] (Fig. 1).

Beyond VHL, a growing body of preclinical research indicates that other genomic and metabolic alterations can also induce a pseudo-hypoxic phenotype, particularly disruptions in the TCA cycle. Inactivation of genes encoding key enzymes of this metabolic pathway, such as succinate dehydrogenase (SDH), fumarate hydratase (FH) and

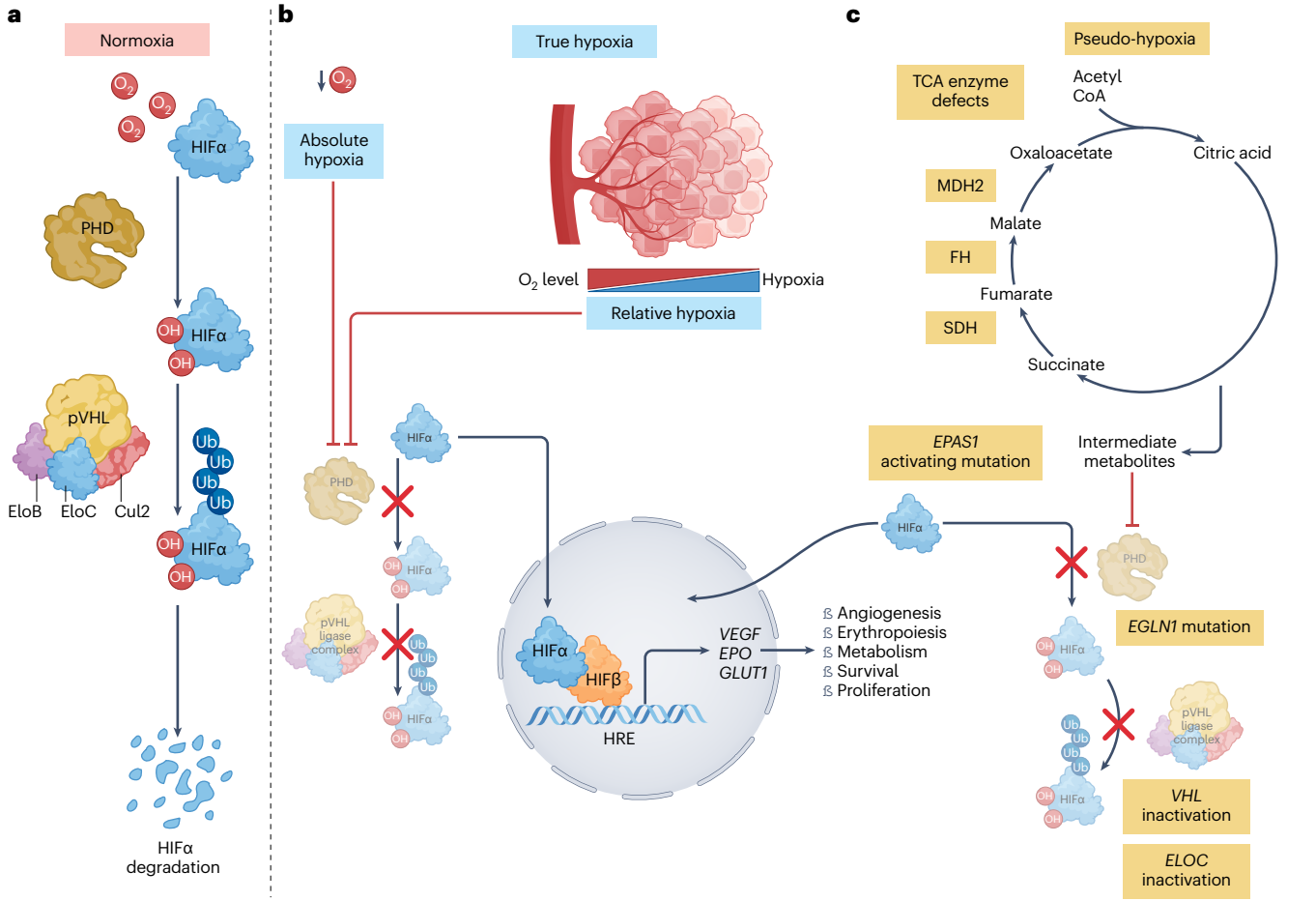


Fig. 1 | Pathways of cellular response to oxygen levels in normoxic, pseudo-hypoxic and true hypoxic conditions. a, Under normoxic conditions, various prolyl hydroxylase domain-containing proteins (PHDs) can use readily available oxygen (O_2) to hydroxylate two crucial regulatory proline residues in hypoxia-inducible factor α-subunits (HIFα; that is, HIF1α and HIF2α). These hydroxylated proline residues enable von Hippel–Lindau disease tumour-suppressor protein (pVHL), the substrate recognition component of an E3 ubiquitin ligase complex, to bind to HIFα and target it for ubiquitination (Ub) and subsequent proteasomal degradation, thereby suppressing HIF transcriptional activity. **b**, Under conditions of relative or true hypoxia, the limited abundance of O_2 restricts PHD activity, which enables HIFα to escape recognition by pVHL, dimerize with HIFβ and subsequently bind to hypoxia-response elements (HREs) in HIF target genes, for example, *VEGF* (encoding vascular endothelial growth factor), *EPO* (encoding erythropoietin) and *GLUT1* (encoding glucose transporter type 1), to drive adaptive cellular responses. **c**, Certain oncogenic alterations induce a 'pseudo-hypoxia' response by upregulating HIF activity despite the presence of adequate levels of O_2, thereby promoting cancer cell survival and proliferation, as well as tumour angiogenesis. These alterations include activating mutations in *EPAS1* (encoding HIF2α); loss-of-function alterations in *VHL* and *ELOC* (encoding elongin C, a member of the VHL E3 ubiquitin ligase complex) or PHD-encoding genes such as *EGLN1*; and various tricarboxylic acid cycle (TCA) enzyme aberrations that result in an accumulation of metabolites that inhibit PHDs (such as succinate and fumarate). Acetyl-CoA, acetyl coenzyme A; FH, fumarate hydratase; MDH2, malate dehydrogenase 2; SDH, succinate dehydrogenase.

malate dehydrogenase 2 (MDH2), leads to an accumulation of intermediate metabolites (in particular, succinate and/or fumarate)[28–30]; these metabolites can outcompete 2-oxoglutarate (also known as α-ketoglutarate), a cofactor required by PHD/EGLN enzymes[31], thereby impairing the hydroxylation of HIFα subunits and promoting their stabilization (Fig. 1). Both germline and somatic mutations in these TCA-cycle genes have been reported as driver or commonly occurring mutations in a variety of cancers.

Germline mutations in SDH genes (*SDHA*, *SDHB*, *SDHC*, *SDHD* and *SDHAF2*) result in hereditary paraganglioma–pheochromocytoma (HPGL/PCC) syndrome[32]. In preclinical models of SDH deficiency, knockdown of SDH expression using small interfering RNAs (siRNAs) leads to intramitochondrial accumulation of succinate[28]. Succinate is then transported into the cytosol, where it directly inhibits PHD/EGLN enzymes, preventing HIFα subunit hydroxylation and degradation, thereby promoting expression of angiogenic targets such as VEGF[28]. In line with these observations, profiling of paraganglioma specimens acquired from patients with HPGL demonstrates increased accumulation of succinate along with increased expression of HIF1α and *VEGF*[33].

FH inactivation, as occurs at the germline level in hereditary leiomyomatosis and renal cell carcinoma (HLRCC; also known as Reed syndrome)[34], leads to intracellular fumarate accumulation, which inhibits PHD/EGLN and stabilizes HIF1α and HIF2α even in normoxia[35]. In mice,

conditional knockout of *Fh* in the kidneys results in the development of clonal renal cystic lesions with profound fumarate accumulation and marked HIF1α and HIF2α overexpression despite normal oxygen levels, accompanied by constitutive activation of HIF target genes such as *Glut1* and *Vegf*[29]. Similarly, in renal tumours from patients with HLRCC, both HIF1α and HIF2α are overexpressed as a result of the inhibition of PHD/EGLNs by fumarate[29,36]. However, concurrent genetic inactivation of HIF1α — but not HIF2α — paradoxically promotes the development of renal cysts in $Fh^{-/-}$ mice, highlighting possible differential roles of HIF paralogues in tumorigenesis[37].

Germline loss-of-function mutations in *MDH2* were initially uncovered in a HPGL/PCC kindred[30]. MDH2 loss causes malate (and secondarily fumarate) to accumulate and inhibit PHD-mediated HIFα subunit prolyl-hydroxylation[38]. However, the interaction between MDH2 and HIF is more complex, given that MDH2 inhibition has also been shown to negatively affect mitochondrial respiration and thus reduce oxygen consumption, with the resultant accumulation of oxygen leading to increased HIF1α degradation[39].

Other genetic alterations resulting in pseudo-hypoxia include inactivating mutations in the PHD/EGLN enzymes themselves. For example, germline mutations in *EGLN1* (encoding PHD2) lead to HIF activation, overexpression of EPO and hereditary erythrocytosis, and development of paragangliomas[40,41]. Finally, activating mutations can directly occur in *EPAS1*, encoding HIF2α. Such mutations tend to occur at the primary hydroxylation site of HIF2α (P531) or surrounding residues (for example, A530, Y532 or D539), thus disrupting the hydroxylation and subsequent degradation of this protein[42,43]. These mosaic, early postzygotic gain-of-function mutations in *EPAS1* underlie Pacak–Zhuang syndrome, which is characterized by early-onset PPGL, pNETs (notably somatostatinomas), polycythaemia and vascular malformations[42–44]. Elongin C (encoded by *ELOC*) is a core member of the VHL E3 ubiquitin ligase complex[12], and somatic *ELOC* mutations define an indolent subtype of RCC[45]. In parallel, germline *ELOC* variants have been reported in a family with VHL-like syndrome in the absence of detectable alterations in *VHL*, with tumours that are responsive to HIF2α inhibitors[46]. These various genomic alterations all converge on activation of the HIF pathway, which ultimately results in the induction of angiogenesis, erythrocytosis and tumour growth.

Basic research on these genetic syndromes has shed light on the inherent differences between the HIFα subunit paralogues. Although HIF1α and HIF2α share similar structures, they differ in terms of tissue distribution and function. Indeed, HIF1α seems to be ubiquitously expressed, albeit at low levels, whereas the expression of HIF2α is restricted to specific cell types, such as chromaffin cells of the adrenal medulla and extra-adrenal paraganglia, and renal interstitial fibroblast-like cells and endothelial cells[47–49]. Both HIF1α and HIF2α induce the expression of angiogenic factors such as VEGF[50]. On the one hand, HIF1α is responsible for mediating the glycolytic switch observed in hypoxia[51], as well as an overall slowing of anabolic processes and hypoxia-induced growth arrest[52,53]. In this respect, HIF1α promotes a more quiescent cell state that could be considered as being protective in hypoxic conditions. On the other hand, HIF2α promotes cell proliferation via activation of MYC and upregulation of cyclin D1 expression, and is therefore regarded as an oncogenic driver[54,55]. The differential effect of the two paralogues arises from several complementary mechanisms. Genome-wide studies have shown that HIF1α and HIF2α have overlapping yet distinct chromatin-binding patterns[56]. Furthermore, HIF1α and HIF2α possess different N-terminal transactivation domains (N-TADs; located in their ODDs and distinct from their C-TADs) that contribute to target specificity by recruiting different cofactors at select regulatory elements; as such, replacing the N-TAD of HIF2α with the analogous domain from HIF1α is sufficient to confer HIF1α target-gene specificity[57]. The 'HIF switch' constitutes another temporal mechanism whereby the expression of HIF1α predominates during acute hypoxia (lasting <24 h), with HIF2α becoming the dominant isoform during prolonged hypoxia (>24 h)[58]. During extended hypoxia, HIFα proteins negatively regulate HIF1α expression by inducing expression of a natural *HIF1A* antisense transcript, which destabilizes the sense *HIF1A* mRNA while HIF2α levels are relatively maintained[58]. HIF3α is a third HIFα subunit paralogue, the function of which remains less understood. The *HIF3A* locus generates several splice-variant isoforms, many of which lack a transactivation domain and seem to act as competitive dominant-negative inhibitors of HIF1α and HIF2α, whereas others seem to transactivate HIF target genes in a context-dependent manner[59].

The role of the VHL–HIF pathway is not restricted to tumours harbouring the aforementioned (epi)genetic alterations. In fact, the uncontrolled growth of solid tumours, combined with the often structurally abnormal microcirculation and chaotic vascularization within these tumours, commonly results in hypoxic areas with increased levels of HIF[60,61]. Overexpression of HIF, particularly HIF2α, has been detected in multiple cancer types and is associated with more hypoxic tumours and a generally worse prognosis[61,62]. Furthermore, HIF2α seems to regulate and independently promote various processes involved in metastasis beyond the formation of new blood vessels, including the epithelial–mesenchymal transition (EMT), invasiveness, motility and survival of seeding tumour cells, thereby driving a more aggressive tumour phenotype[63–65].

Although HIF2α seems to be a major oncogenic driver in certain tumour types, such as ccRCC (in which it might constitute a vital dependency), evidence indicates that it might also contribute to tumorigenesis and drive the growth of several other cancers[66,67]. For example, short-hairpin RNA (shRNA)-mediated knockdown of HIF2α expression impedes tumour formation and growth in mouse xenograft models of glioblastoma, colorectal cancer and lung cancer[67]. Therefore, HIF2α has emerged as an attractive therapeutic target in a variety of cancer types.

Early development of HIF2α inhibitors

Efforts to develop pharmacological inhibitors of HIF2α emerged in the context of ccRCC. Initially, systemic treatment strategies for ccRCC focused on targeting downstream effectors of HIF such as VEGF. However, direct targeting of HIF2α, the master transcription factor driving these oncogenic changes, was hypothesized to be another, and perhaps more effective, strategy[68]. As a proof of concept, HIF2α inhibition through shRNA-mediated silencing[69] or expression of the dominant-negative HIF3α4 splice variant, which is capable of dimerizing with HIF1β but lacks transactivation domains[70], was shown to suppress the growth of VHL-null ccRCC tumours in mouse models. Early studies also identified small-molecule compounds that could repress HIF2α translation by enhancing the binding of iron regulatory protein 1 (IRP1; also known as aconitate hydratase) to an iron-responsive element (IRE) within the 5′-untranslated region of the *EPAS1* mRNA[71]. However, from a pharmacological standpoint, DNA-binding transcription factors such as HIF2α had long been considered directly 'undruggable' using small-molecule inhibitors[72]. Almost 15 years passed following the initial cloning and characterization of HIF2α in 1997 (ref. 7) before researchers identified a hydrophobic pocket within its PAS-B domain as well as artificial ligands that could bind within this allosteric pocket

and inhibit the dimerization of HIF2α and HIF1β[73–75], paving the way for the development of small-molecule HIF2α inhibitors (Fig. 2).

PT2385

PT2385 was the first oral, selective, small-molecule HIF2α inhibitor to emerge as a lead compound from a structure-based drug discovery programme and showed potent antitumour activity in preclinical studies[76]. PT2385 was also the first HIF2α inhibitor to be tested in humans, in a phase I trial involving 51 patients with heavily pretreated advanced-stage ccRCC, of whom 26 were in the dose-escalation and 25 in the dose-expansion phases[77]. All tested doses of PT2385, ranging from 100 to 1,800 mg twice daily, resulted in rapid decreases in plasma levels of EPO, indicating successful inhibition of the HIF2 signalling pathway[77]. From a clinical standpoint, the objective response rate (ORR) was 14%, with 5 partial responses (PRs) and 1 complete response (CR) among 50 evaluable patients[77]. Notably, the drug was deemed safe and well tolerated, with no dose-limiting toxicities. The most common grade ≥3 adverse events (AEs) were anaemia (in 5 patients, 10%), which was asymptomatic and in line with the reported decreases in EPO levels, and hypoxia (also in 10%, all grade 3), which led to treatment interruption in 2 patients. The activity of PT2385 was restricted by its pharmacokinetic properties, notably its globally high and patient-dependent rates of glucuronidation, resulting in rapid drug clearance (mean half-life of 17 h) and thus substantial underexposure[77,78].

PT2399

As part of the same drug discovery effort that identified PT2385, the closely related tool compound PT2399 was developed and advanced to preclinical studies[79,80]. PT2399 is a selective HIF2α inhibitor, with no effect on HIF1α despite its structural similarity to HIF2α. PT2399 resulted in downregulation of HIF2-dependent genes, such as *VEGFA*, *CCND1*, *SLC2A1* and *NDRG1*, in ccRCC cells, and halted their growth in soft agar colonies and in vivo in subcutaneous and orthotopic tumour models in mice (including patient-derived xenografts)[79,80]. Notably, sensitivity to PT2399 was variable across ccRCC cell lines regardless of *VHL* mutational status[79]. However, less-favourable pharmacokinetics, in part related to metabolic clearance and drug–drug interaction liabilities, limited further development of PT2399, and this compound was ultimately not tested clinically, being superseded by improved analogues such as PT2385 and, later, PT2977.

PT2977

PT2977 (also known as MK-6482)[81] was developed as a regioisomer of PT2385, with a geminal difluoro (-CF_2-) group replaced by a vicinal difluoro (-CHF-CHF-) moiety, which resulted in reduced phase II metabolic conjugation (particularly glucuronidation), decreased lipophilicity and increased binding affinity, subsequently enhancing its potency tenfold compared to PT2385 in preclinical mouse models[78]. As a result, PT2977/MK-6482 – subsequently named belzutifan – emerged as the leading HIF2α inhibitor for clinical testing.

State of the art in HIF2α inhibition and current clinical applications

Belzutifan is the first-in-class and current state-of-the-art HIF2α inhibitor. In a first-in-human phase I dose-escalation study (part of the LITESPARK-001 trial; Table 1) involving 43 patients with various advanced-stage solid tumours, belzutifan had a favourable pharmacokinetic and pharmacodynamic profile, and was well tolerated at once-daily doses ranging from 20 to 160 mg; plasma EPO levels declined in all patients, with maximal suppression achieved at doses ≥120 mg, matching the levels seen with high exposure to PT2385 (800 mg twice daily)[78,82,83]. In this cohort, six PRs were observed with doses ≥120 mg, five in patients with RCC and one with anaplastic ependymoma[83]. Subsequently, a phase II dose-expansion portion was performed in a cohort of 55 patients with pretreated advanced-stage ccRCC using the recommended phase II dose of 120 mg once daily. In this cohort, the ORR was 25% (PRs in 14 of 55 patients), and 67% of patients (35 of 52 evaluable) had tumour shrinkage compared to baseline measurements[83,84]. Consistent with the toxicity profile of PT2385, the most common grade 3 AEs in the dose-expansion cohort were anaemia (in 27% of patients) and hypoxia (16%), albeit occurring at slightly higher rates than those observed with PT2385; no grade 4

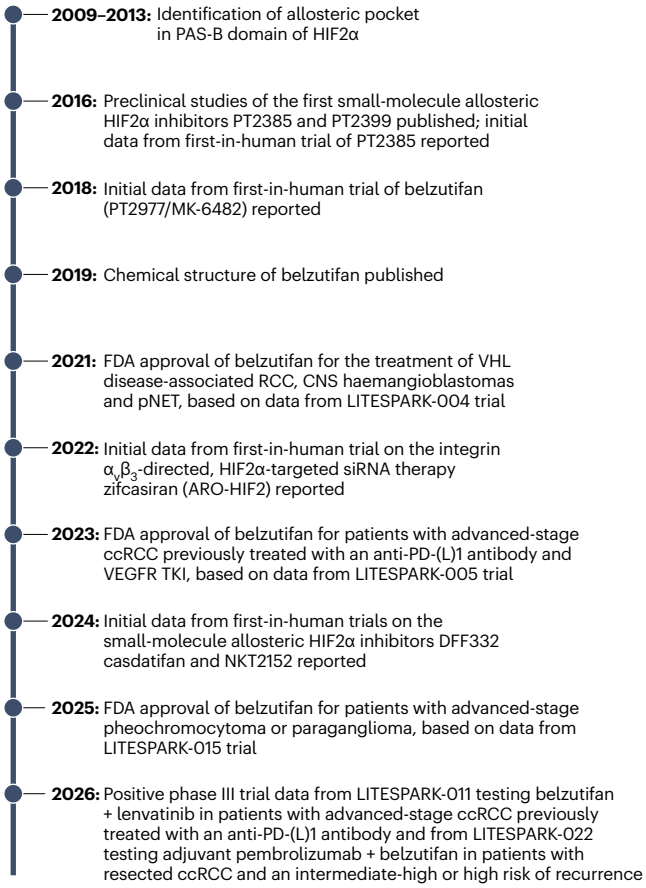


Fig. 2 | Timeline of key landmarks in the development of HIF2α antagonists for the treatment of cancer. The initial discovery of a druggable allosteric pocket in the Per-ARNT-Sim (PAS)-B domain of hypoxia-inducible factor 2α (HIF2α) in 2009 enabled the development of the first small-molecule inhibitors of this transcription factor, followed within a decade by early-phase clinical trials testing the HIF2α inhibitors PT2385 and belzutifan. Multiple subsequent approvals of belzutifan and the emergence of novel small-molecule inhibitors and RNA-based targeting strategies highlight the rapid clinical translation of HIF2α inhibition across both sporadic and von Hippel–Lindau (VHL)-associated renal cell carcinoma (RCC) and other VHL-associated malignancies. ccRCC, clear-cell renal cell carcinoma; CNS, central nervous system; pNET, pancreatic neuroendocrine tumour; siRNA, small interfering RNA; TKI, tyrosine kinase inhibitor; VEGFR, vascular endothelial growth factor receptor.

Table 1 | Completed or reported clinical trials of HIF2α inhibitors across RCC and other malignancies

Trial name (phase); Refs.	Experimental treatment (drug targets)	Primary end points	Setting; line of therapy (*n*)	ORR	Other outcomes and comments	Common grade ≥3 AEs
PT2385-102 (I); NCT02553356	PT2385 (HIF2α)	PK, PD	Healthy volunteers (16)	NA	–	–
PT2385-101/ MK-3795-001 (I)[77,120]	PT2385	MTD, RP2D	ccRCC; previously treated (51)	14%	–	TEAEs: 33% Anaemia: 10% Hypoxia: 10%
	PT2385+nivolumab (PD-1)	MTD, RP2D	ccRCC; previously treated (50)	22%	–	Anaemia: 4% Hypoxia: 4%
	PT2385+cabozantinib (VEGFR1–3, MET, AXL)	MTD, RP2D	ccRCC; previously treated (9)	NA	Discontinued owing to development of belzutifan (NCT02293980)	–
PT2385-202/ MK-3795-003 (II); NCT03108066	PT2385	ORR	VHL-associated ccRCC; first line (4)	0%	SD in all 4 patients	–
ABTC 1602 (II)[176]	PT2385	ORR	Glioblastoma; recurrent, no prior anti-angiogenic therapies (24)	0%	mPFS: 1.8 months mOS: 7.7 months	Hypoxia: 8% Anaemia: 4%
LITESPARK-001 (I/II)[83,84,177]	Belzutifan (HIF2α)	MTD and/or RP2D	ccRCC; any line (55)	24%	mPFS: 14.5 months	TRAEs: 40% Anaemia: 24% Hypoxia: 13%
			Glioblastoma; first recurrence after CRT (25)	0%	mPFS: 1.4 months	AEs: 60% Any-grade anaemia: 64%
LITESPARK-004 (II)[85,86,140]	Belzutifan	ORR	VHL-associated ccRCC; first line (61)	70.5% (5-year follow-up)	mPFS: 49.8 months mOS: NR (mFU 49.9 months)	TRAEs: 18% Anaemia: 11% Hypoxia: 2%
LITESPARK-005 (III)[90]	Belzutifan vs everolimus (mTOR)	PFS, OS	ccRCC; previously treated (736)	21.9% vs 3.5%	mPFS: 5.6 months in both arms 18-month PFS: 24.0% vs 8.3% mOS: 21.4 vs 18.1 months (HR 0.88, 95% CI 0.73–1.07; *P*=0.20)	AEs: 62% Anaemia: 33% Hypoxia: 11%
LITESPARK-013 (II)[91]	Belzutifan 120 mg vs 200 mg	ORR	ccRCC; previously treated (154)	23.7% vs 23.1%	mPFS: 7.3 vs 9.1 months (HR 0.94, 95% CI 0.63–1.4; *P*=0.39) mOS: NR in either group (mFU: 20.1 months)	AEs: 46% with both doses Anaemia: 20% vs 27% Hypoxia: 21% vs 22%
LITESPARK-003 (II)[105]	Belzutifan+cabozantinib	ORR	ccRCC; first line (50)	70%	mPFS: 30.3 months mOS: NR (mFU: 34.4 months)	AEs: 60% Anaemia: 10% Hypoxia: 6%
			ccRCC; previously treated (52)	30.8%	mPFS: 13.8 months mOS: 26.7 months	AEs: 73% Anaemia: 15% Hypoxia: 4%
LITESPARK-010 (I)[109,110]	Belzutifan+lenvatinib (VEGFR1–3, KIT, FGFR1–4, PDGFRα, RET)	PK, safety	ccRCC; previously treated (23)	50%	mPFS: 13.7 months mOS: NR (mFU 14.4 months)	TRAEs: 71% Anaemia: 29% Hypoxia: 4%
LITESPARK-011 (III)[108]	Belzutifan+lenvatinib vs cabozantinib	PFS	ccRCC; any line, after anti-PD-(L)1 antibody-based therapy (708)	52.6% vs 40.2%	mPFS: 14.8 vs 10.7 months (HR 0.70, 95% CI 0.59–0.84; *P*=0.00007) mOS: 34.9 vs 27.6 months (HR 0.85, 95% CI 0.68–1.05)	TEAEs: 84.1% vs 82.7% (5.4% vs 3.2% grade 5)
LITESPARK-022 (III)[123]	Belzutifan+pembrolizumab (PD-1) vs pembrolizumab alone	DFS	Resectable ccRCC; adjuvant setting after nephrectomy/ metastatectomy (1,841)	NA	mDFS: NR in either arm (mFU 28.4 months) mDFS at 2 years: 80.7% vs 73.7% (HR 0.72, 95% CI 0.59–0.87; *P*= 0.0003) OS data immature (HR 0.78, 95% CI 0.51–1.19; *P*=0.12)	TEAEs: 52.1% vs 30.2% Anaemia: 12.1% vs 0.4% Increased ALT: 6.4% vs 2.0% Hypoxia: 4.6% vs 0%

Table 1 (continued) | Completed or reported clinical trials of HIF2α inhibitors across RCC and other malignancies

Trial name (phase); Refs.	Experimental treatment (drug targets)	Primary end points	Setting; line of therapy (*n*)	ORR	Other outcomes and comments	Common grade ≥3 AEs
LITESPARK-024 (I/II)[131]	Belzutifan + 75/100/125 mg palbociclib (CDK4/6)	Safety, RP2D, ORR	ccRCC; after at least 1 VEGFR TKI and 1 ICI (58; 20/19/19)	15.0%/0%/21.1%	mPFS: 7.2/5.4/9.1 months	TRAEs: 65.0%/78.9%/78.9% Anaemia: 50.0%/52.6%/57.9%
KEYMAKER-U03 Substudy 03A (I/II)[111]	Belzutifan + pembrolizumab + lenvatinib	ORR, safety	ccRCC; first line (90)	77.5%	mPFS: 31.8 months	TRAEs: 70% Anaemia: 76%
	Belzutifan + pembrolizumab + vibostolimab (TIGIT)	ORR, safety	ccRCC; first line (90)	42.5%	mPFS: 15.2 months	TRAEs: 69% Anaemia: 81%
KEYMAKER-U03 Substudy 03B (I/II)[106]	Belzutifan + lenvatinib	ORR, safety	ccRCC; previously treated (64)	50%	mPFS: 15.0 months mOS: 36.4 months	TRAEs: 63%
	Belzutifan + pembrolizumab	ORR, safety	ccRCC; previously treated (62)	23%	mPFS: 5.5 months mOS: 22.9 months	TRAEs: 45%
LITESPARK-015 (II)[89,167,183]	Belzutifan	ORR	VHL-associated tumours (44)	RCC (*n*=34) 88.2%; solid/cystic CNS-HB (*n*=22) 59.1%; solid CNS-HB (*n*=16) 81.3%; pNET (*n*=21) 90.5%	No PFS or OS events had occurred (mFU: 14.9 months)	AEs: 39%
			PPGL (72)	26%	mPFS: 22.3 months mOS: NR (mFU 30.2 months)	AEs: 46% Anaemia: 22% Hypoxia: 10%
			Sporadic pNETs (70)	10%	mPFS 3.9 months mOS: NR (mFU 30.9 months)	TRAEs: 16% Any-grade anaemia: 73%
ARC-20 (I/Ib)[94,116,117]	Casdatifan (HIF2α)	Safety, tolerability, ORR	ccRCC; previously treated (127; 32 in 100 mg QD cohort)	31% (35% with 100 mg QD)	mPFS: 12.2 months (NR with 100 mg QD at mFU of 12 months)	Dose reductions owing to TEAEs: 24% (22% with 100 mg QD); due to anaemia in 14% (9%) and hypoxia in 7% (3%)
	Casdatifan + cabozantinib	Safety, tolerability, ORR	ccRCC; previously treated (42, 24 efficacy evaluable)	46%	mFU only: 5.3 months	TRAEs: 48% Anaemia: 24% Hypoxia: 7%
NKT2152-101 (I/II)[97]	Imdatifan (HIF2α)	DLTs, RDE, RP2D, ORR	ccRCC; previously treated (96)	24%	mPFS: 7.5 months Without prior mTOR inhibitor (*n*=35): ORR 40%, mPFS, 9.4 months	TRAEs: 32% Anaemia: 15% Hypoxia: 16%
CDFF332A12101 (I/Ib)[95]	DFF332 (HIF2α)	PK, PD	ccRCC; previously treated (40)	5%	–	TRAEs: 5% Anaemia: 0% Hypoxia: 0%
	DFF332 + spartalizumab (PD-1) + taminadenant (A2AR)	PK, PD	ccRCC; previously treated (NA)	NA	Discontinued (NCT04895748)	–
	DFF332 + everolimus	PK, PD	ccRCC; previously treated (NA)	NA	Discontinued (NCT04895748)	–
AROHIF21001 (Ib)[104]	Zifcasiran (HIF2α)	RP2D, safety, tolerability	ccRCC; previously treated (26)	7.7%	–	TRAEs: 15% Hypoxia: 8% Neurotoxicity: 23%

A2AR, adenosine A_{2A} receptor; AEs, adverse events; ccRCC, clear-cell renal cell carcinoma; CDK4/6, cyclin-dependent kinases 4 and 6; CNS-HB, central nervous system haemangioblastoma; CRT, chemoradiotherapy; DFS, disease-free survival; DLTs, dose-limiting toxicities; FGFR1–4, fibroblast growth factor receptors 1–4; HIF2α, hypoxia-inducible factor 2α; ICI, immune-checkpoint inhibitor; mDFS, median disease-free survival; mFU, median follow-up; mOS, median overall survival; mPFS, median progression-free survival; MTD, maximum tolerated dose; NA, not applicable/available; NR, not reached; ORR, objective response rate; OS, overall survival; PD, pharmacodynamics; PDGFRα, platelet-derived growth factor receptor-α; PFS, progression-free survival; PK, pharmacokinetics; pNETs, pancreatic neuroendocrine tumours; PPGL, pheochromocytoma and/or paraganglioma; QD, once daily; RCC, renal cell carcinoma; RDE, recommended dose for expansion; RP2D, recommended phase II dose; SD, stable disease; TEAEs, treatment-emergent adverse events; TKI, tyrosine kinase inhibitor; TRAEs, treatment-related adverse events; VEGFR1–3, vascular endothelial growth factor receptors 1–3; VHL, von Hippel–Lindau.

or 5 AEs were reported, and only 2 patients discontinued treatment owing to AEs, namely hypoxia[83,84].

These encouraging initial results led to a phase II trial testing belzutifan in treatment-naive patients with VHL disease-associated RCC (LITESPARK-004)[85]. A total of 61 patients with germline pathogenic variants of *VHL* and at least one measurable RCC tumour were enrolled and received belzutifan at a dose of 120 mg once daily[85]. At the 5-year analysis, 57 out of 61 patients (93%) had measurable RCC tumour shrinkage, including 7 patients (11%) with a CR and 36 (59%) with a PR as best response, amounting to an ORR of 70.5%[86]. After at least 60 months of follow-up in all patients, 35 were still on treatment, with a median treatment duration of 61.8 months[86]. Non-RCC VHL-associated tumours present in the 61 enrolled patients were also evaluated in LITESPARK-004, with clinically meaningful activity observed across multiple disease sites. Among 50 patients with evaluable CNS haemangioblastomas, the ORR was 50% (6 CRs and 19 PRs) when all baseline lesions, including cystic components, were considered in analysis performed at a median follow-up of 61.8 months[87]. In a previous analysis at a median follow-up of 38.0 months, when data were restricted to the 25 patients with at least one measurable solid CNS haemangioblastoma lesion, the ORR was 76% (1 CR and 18 PRs)[87]. All 61 patients enrolled in LITESPARK-004 also had VHL-associated pancreatic lesions, including pNETs in 22 patients, with an ORR per RECIST version 1.1 of 84% for all pancreatic lesions (including 17 CRs) and 91% for pNETs (including 7 CRs) reported in an analysis at a median follow-up of 37.8 months[88]. At the 5-year follow-up analysis, the ORR remained very high at 90% among 20 evaluable patients with pNETs, including 6 CRs[86]. In the 5-year update, all 14 evaluable patients with active retinal haemangioblastoma had improvements, yielding a 100% ophthalmological response rate across 18 affected eyes on ophthalmological assessment[86]. More importantly, the number of surgical procedures decreased substantially after initiation of belzutifan: 75% of patients ($n = 46$) underwent 97 VHL-related procedures in the 5 years preceding the start of belzutifan treatment, whereas 31% of patients ($n = 19$) underwent 24 surgical procedures, mostly for RCC tumours, after receiving this agent. In terms of toxicity, grade 3 treatment-related AEs (TRAEs) occurred in 18% of patients ($n = 11$), with anaemia representing the most common severe toxicity (11%, $n = 7$); no grade 4 or 5 AEs were attributed to belzutifan. Dose reductions were required in 11 patients (18%), and treatment discontinuation owing to AEs was uncommon ($n = 2$; 3%)[86]. Based on earlier results from LITESPARK-004, in August 2021, belzutifan was granted FDA approval for the treatment of adult patients with VHL disease-associated RCC, CNS haemangioblastoma and/or pNET not requiring immediate surgery[2] (Fig. 2). The efficacy of belzutifan in VHL disease-associated tumours was also confirmed in LITESPARK-015, an ongoing, open-label, multicohort phase II trial (NCT04924075)[89] evaluating belzutifan in patients with various tumour types. Cohort B1 enrolled 44 patients from China or Japan with VHL disease-associated localized tumours[89]. In this cohort, the primary end point was met, with ORR of 88.2% for VHL-associated RCC ($n = 34$)[89]. ORRs were 90.5% for pNETs ($n = 21$), 59.1% for solid and cystic CNS haemangioblastomas ($n = 22$), and 81.3% for solid CNS haemangioblastomas ($n = 16$), whereas the ORR for PPGL ($n = 6$) was substantially lower, at 16.7%[89]. Median duration of response (DOR) and progression-free survival (PFS) were not reached, with 24-month PFS rates of 88.1% for RCC, 88.4% for pNETs, 90.4% for solid and cystic CNS haemangioblastomas, and 100% for both solid CNS haemangioblastomas and PPGL[89].

Building upon this experience, the efficacy of belzutifan was investigated in patients with sporadic ccRCC. In the phase III LITESPARK-005 trial, 746 patients with advanced-stage ccRCC previously treated with an immune-checkpoint inhibitor (ICI; specifically, an anti-PD-(L)1 antibody) and a VEGFR tyrosine kinase inhibitor (TKI) were randomly assigned (1:1) to receive either belzutifan 120 mg or everolimus 10 mg once daily[90]. PFS and overall survival (OS) were the co-primary end points, and crossover was not planned[90]. At the first interim analysis at 18 months, 24.0% of patients in the belzutifan group and 8.3% in the everolimus group were alive without disease progression ($P = 0.002$), meeting the PFS end point[90]. The ORR was similarly higher with belzutifan compared to everolimus (21.9% versus 3.5%; $P < 0.001$)[90]. It is important to note that the response rates observed in LITESPARK-004 reflect treatment-naive, mostly localized RCC lesions in germline *VHL* mutation carriers, whereas LITESPARK-005 enrolled patients with heavily pretreated metastatic sporadic ccRCC; therefore, ORR comparisons across these settings should not be interpreted as only reflecting intrinsic differences in HIF2α dependence. With regard to toxicities, the incidence of grade ≥3 AEs of any cause was numerically similar between the two arms (61.8% with belzutifan and 62.5% with everolimus)[90]. The most common grade ≥3 AEs with belzutifan were anaemia and hypoxia, and with everolimus they were anaemia, hyperglycaemia and hypertriglyceridaemia[90]. Despite the significant improvement in PFS, the co-primary OS end point was not met at the second interim analysis with a median follow-up duration of 25.7 months, with a median OS of 21.4 months with belzutifan versus 18.1 months with everolimus[90] (HR 0.88, 95% CI 0.73–1.07; $P = 0.20$). This disconnect might, at least in part, reflect post-progression therapies attenuating OS differences, considering that 64.5% of patients in the everolimus arm received subsequent therapies, mostly VEGFR TKIs, compared to only 47.3% in the everolimus arm. Based on the improvement in PFS, in December 2023, belzutifan was granted FDA approval for the treatment of patients with advanced-stage RCC following disease progression on a prior anti-PD-(L)1 antibody and a VEGFR TKI[3] (Fig. 2).

Subsequently, results from the LITESPARK-013 dose-finding trial further supported 120 mg as the optimal dose of belzutifan for patients with previously treated advanced-stage ccRCC[91]. In this phase II trial, 154 patients were randomly assigned (1:1) to receive belzutifan 120 mg or 200 mg once daily, with similar efficacy observed in both arms in terms of ORR (23.7% and 23.1%, respectively; $P = 0.53$), PFS (HR 0.94, 95% CI 0.63–1.40) and OS (HR 1.11, 95% CI 0.65–1.90)[91]. Overall, the incidence of grade 3–4 TRAEs was almost identical with the 120 mg and 200 mg doses (46.1% versus 46.2%), although several toxicities occurred more often at the higher dose; treatment discontinuation owing to AEs (5.3% versus 14.1%) and dose modifications (46.1% versus 57.7%) were also more common in the 200 mg arm[91]. In particular, anaemia was less frequent and generally less severe with belzutifan 120 mg (any grade 75.0% versus 80.8%; grade 3–4 19.7% versus 26.9%), translating into fewer dose interruptions (2.6% versus 6.4%) and a reduced need for erythropoiesis-stimulating agents (ESAs) plus red blood cell transfusion (7.5% versus 17.1%)[91]. Hypoxia occurred at similar rates in both arms (any grade 23.7% versus 26.9%; grade 3–4 21.1% versus 21.8%), albeit less frequently necessitating dose interruptions in the 120 mg arm (5.3% versus 9.0%).

This body of evidence established belzutifan as the first HIF2α inhibitor approved for cancer therapy in the context of both heavily pretreated ccRCC and treatment-naive VHL disease-associated tumours. Building on this success, a growing body of clinical trial data is further defining the role of HIF2α inhibition in the treatment of these tumour types (Table 1), while ongoing trials are evaluating novel inhibitors and rational combination regimens in different disease settings

(Supplementary Table 1). Together, these datasets provide a framework for assessing current progress and contextualizing the development of emerging agents.

Other HIF2α inhibitors entering clinical development

Casdatifan

Casdatifan (also known as AB521) is another orally bioavailable, small-molecule, allosteric inhibitor of HIF2α. Similar to belzutifan, casdatifan contains a vicinal difluoride (-CHF-CHF-) group; however, in casdatifan, this feature is present in a *cis*-difluorinated tetralin core, in contrast to the *trans* configuration in belzutifan[92]. The *cis* arrangement has been associated with greater facial polarization, a property that might affect binding interactions within the HIF2α PAS-B pocket[92]. Pharmacologically, casdatifan has a reported terminal half-life of approximately 24 h, compared with about 14 h for belzutifan, potentially supporting more stable systemic exposure[93]. In contrast to belzutifan, which is predominantly hepatically metabolized, approximately 30% of casdatifan molecules are renally cleared[93]. In pharmacokinetics and pharmacodynamics analyses, a 20 mg once-daily dose of casdatifan induces an approximately 70% reduction in plasma EPO levels, a magnitude similar to that observed with belzutifan 120 mg once daily[93]. In addition, casdatifan exhibits linear pharmacokinetics, enabling further increases in plasma exposure with dose escalation and a greater reduction in EPO levels (mean reduction of 85% at the 100 mg once-daily dose)[93]. In the phase I–Ib ARC-20 trial, casdatifan monotherapy demonstrated encouraging antitumour activity in patients with HIF2α inhibitor-naive advanced-stage sporadic ccRCC previously treated with an anti-PD-(L)1 antibody and a VEGFR TKI (median three prior systemic lines)[94]. Among 127 patients treated across monotherapy cohorts of 50 mg once daily, 50 twice daily, 100 mg once daily and 150 mg once daily, the confirmed ORR was 31% overall, including one CR, and 35% (all PRs) in the 100 mg once-daily cohort ($n = 32$)[94]. At a median follow-up of 12 months in the 100 mg once-daily cohort and 15 months in the overall population, median PFS was 12.2 months and not reached, respectively[94]. Casdatifan had a manageable safety profile, with dose reductions due to treatment-emergent AEs (TEAEs) occurring in 24% of patients overall (22% at 100 mg once daily), most commonly owing to anaemia (14% and 9%, respectively) and hypoxia (7% and 3%, respectively)[94]. Treatment discontinuation owing to TRAEs occurred in only 3% of patients, and no treatment-related deaths were reported[94]. Notably, the dose-expansion phase of this ongoing trial also includes arms testing casdatifan in combination with cabozantinib or zimberelimab or zimberelimab and ipilimumab in patients with ccRCC (NCT05536141; Supplementary Table 1). Casdatifan is also being evaluated in the neoadjuvant setting Pre-NEOSHIFT-RCC trial (NCT07397611), a randomized, open-label phase II trial evaluating casdatifan alone or in combination with zimberelimab in patients with resectable ccRCC, with the primary objective of assessing tumour size reduction prior to nephrectomy.

DFF332

DFF332 is also an orally bioavailable, small-molecule, allosteric HIF2α inhibitor that disrupts HIF2α–HIF1β dimerization. In a phase I dose-escalation trial (NCT04895748), 40 patients with advanced-stage sporadic ccRCC received DFF332 at doses ranging from 50 mg weekly to 150 mg daily[95]. Patients were heavily pretreated, with 87.5% having received at least two prior lines of therapy[95]. Pharmacokinetic data demonstrated rapid oral absorption of DFF332, with a median time to peak drug concentration of 1–2 h, and very slow elimination, with an estimated effective half-life of ~85 days[95]. A favourable safety profile was reported, with treatment-related anaemia in only 5 patients (13%) and no hypoxia[95]. At data cut-off, 2 of 40 patients (5%) had a PR, with no CRs reported[95]. The study sponsor subsequently discontinued further evaluation of DFF332.

Imdatifan

Imdatifan (also known as NKT2152 or HS-10516) is another small-molecule oral allosteric inhibitor of HIF2α. Given its long half-life of approximately 38 days[96], imdatifan was explored at several dose levels in a first-in-human phase I–II trial involving 96 patients with previously treated advanced-stage sporadic ccRCC[97]. Initial dose escalation evaluated dosages of 50–300 mg once daily, which was later refined to a loading phase of 100 or 200 mg daily for 7, 14 or 28 days, followed by maintenance dosing of 50 mg daily or 100 or 200 mg weekly to limit drug accumulation while maintaining therapeutic exposure[97]. Grade ≥3 TRAEs occurred in 32% of patients (7.7–67% across dose cohorts), including anaemia in 15% (0–33%) and hypoxia in 16% (0–33%). At a median follow-up duration of 12 months, the ORR by RECIST 1.1 among 81 efficacy-evaluable patients was 24% and median PFS was 7.5 months[97]. In 35 patients who had not received prior treatment with an mTOR inhibitor, the ORR was 40% and median PFS was 9.4 months. However, this trial has been terminated owing to sponsor portfolio reprioritization (NCT05119335).

BPI-452080

BPI-452080 — yet another orally bioavailable, selective, small-molecule, allosteric HIF2α inhibitor — demonstrated dose-dependent tumour growth inhibition in various preclinical models of RCC as well as favourable bioavailability and tolerability in multiple preclinical species[98]. A phase I trial of this agent has been initiated in China (NCT05843305) and will include patients with advanced-stage ccRCC, including VHL disease-associated RCC, as well as other solid tumours (Supplementary Table 1).

KD061

KD061 is a novel, first-in-class, orally available ferroptosis inducer that targets iron–sulfur cluster assembly 2 homologue (ISCA2)[99]. Ferroptosis is an iron-mediated form of regulated cell death characterized by the accumulation of lipid peroxides, and has gained attention as a possible vulnerability in RCC given its lipid-rich nature[100]. In addition to inducing ferroptosis, disrupting mitochondrial iron–sulfur cluster assembly via inhibition ISCA2 can also result in downregulation of HIF2α protein expression via IRE-dependent translational repression[99]. KD061 demonstrated robust in vitro activity across VHL-deficient ccRCC cell lines, with a 92-cell-line screen beyond ccRCC suggesting additional tumour types sensitive to ISCA2 inhibition (including melanoma and clear-cell ovarian cancer), and resulted in >65% tumour growth inhibition in mouse models[99]. With a dual mechanism of HIF2α repression and ferroptosis induction[99], KD061 offers a mechanistically distinct and potentially complementary alternative to canonical HIF2α inhibitors such as belzutifan.

HIF2α siRNA

RNA interference (RNAi)-based strategies targeting HIF2α have produced promising results in preclinical models of ccRCC. In xenograft models, a HIF2α siRNA, conjugated to a ligand for $\alpha_v\beta_3$ and $\alpha_v\beta_5$ integrins, which are frequently expressed in ccRCC, resulted in

dose-dependent knockdown of *EPAS1* mRNA, with the extent of suppression correlating with tumour growth inhibition[101,102]. Similar results were observed in patient-derived ccRCC xenograft models, in which treatment with the siRNA reduced HIF2α levels, downregulated the expression of key HIF2-target genes, including *VEGF*, and demonstrated comparable antitumour activity to that of the small-molecule HIF2α inhibitor PT2399 (ref. 103). Notably, the siRNA construct was capable of effectively depleting both wild-type and mutant forms of HIF2α, including those resistant to small-molecule inhibitors.

These development efforts lead to clinical testing of zifcasiran (also known as ARO-HIF2 and ADS-007), a chemically stabilized, double-stranded siRNA trigger — designed to engage the endogenous RNAi machinery to selectively degrade *EPAS1* mRNA — conjugated to an $\alpha_v\beta_3$ ligand that facilitates uptake by cancer cells; thus, this agent presents a mechanistically distinct alternative to small-molecule HIF2α inhibitors[104]. AROHIF21001 was a multicentre, open-label, dose-finding phase I trial evaluating zifcasiran in patients with advanced-stage ccRCC with progressive disease after two or more prior lines of therapy, including a VEGFR TKI and an ICI (Table 1). In total, 26 patients received intravenous zifcasiran at weekly doses of 225 mg ($n = 7$), 525 mg ($n = 10$) or 1,050 mg ($n = 9$)[104]. *EPAS1* mRNA and/or HIF2α protein levels decreased between pre-treatment and post-treatment tumour biopsy samples in most patients evaluated[104]. However, an ORR of only 7.7% together with off-target neurotoxicity led to discontinuation of further clinical development[104].

Role for combinations

Despite promising antitumour efficacy, HIF2α inhibition alone does not fully suppress the adaptive programmes driven by hypoxia. Combinatorial strategies, supported by both preclinical models and emerging clinical data (Table 1 and Supplementary Table 1), thus aim to exploit mechanistic synergies and counteract resistance.

VEGFR TKIs

The combination of HIF2α inhibitors and VEGFR TKIs offers a complementary approach to limiting tumour growth by targeting the HIF2–VEGF axis at two distinct levels: inhibition of HIF2α suppresses *VEGF* transcription and VEGFR TKIs mitigate downstream VEGF signalling. This two-pronged approach might enhance antitumour efficacy and help to overcome resistance mechanisms; therefore, several trials have explored the potential of this combinatorial strategy.

The phase II LITESPARK-003 trial evaluated the combination of belzutifan (120 mg once daily) and cabozantinib (60 mg once daily) in patients with advanced-stage ccRCC across two cohorts: treatment-naive patients (cohort 1) and patients previously treated with ICIs (cohort 2)[105]. In cohort 1, after a median follow-up of 34.4 months, 35 of 50 patients (70%) had a confirmed objective response, including 6 CRs, with a median PFS of 30.3 months and a DOR of 29.1 months[105]. In cohort 2, with a median follow-up duration of 49.9 months, 16 of 52 patients (30.8%) had an objective response (including 2 CRs), with a similar ORR regardless of prior treatment with VEGFR TKIs (29%, compared with 32% in the ICI-only subgroup); the median PFS in this cohort was 13.8 months but the median DOR was 30.4 months[105]. Median OS was not reached in cohort 1 and was 26.7 months in cohort 2 (ref. 105).

In the phase I–II KEYMAKER-U03 Substudy 03B, the combination of lenvatinib (20 mg once daily) and belzutifan (120 mg once daily) demonstrated encouraging clinical activity in patients with previously treated advanced-stage ccRCC[106]. Among 64 heavily pretreated patients (median three prior lines of treatment), the combination yielded an ORR of 50% (including three CRs)[106]. At a median follow-up of 22.9 months, median PFS was 15.0 months, median DOR was not reached (range 2.9–38.4+ months) and median OS was 36.4 months[106]. However, this combination was also associated with a high incidence of grade ≥3 AEs (63%)[106]. Nevertheless, the promising efficacy data provided the rationale for the ongoing phase III LITESPARK-011 trial (NCT04586231), which is evaluating whether belzutifan–lenvatinib improves PFS and OS compared to cabozantinib monotherapy in patients with advanced-stage ccRCC that has progressed following anti-PD-(L)1 antibody therapy[107] (Table 1). At the second interim analysis, the combination significantly improved PFS compared with cabozantinib (median 14.8 months versus 10.7 months; HR 0.70, 95% CI 0.59–0.84; $P = 0.00007$) and resulted in a higher ORR (52.6% versus 40.2%). Although OS numerically favoured the combination (median 34.9 months versus 27.6 months; HR 0.85, 95% CI 0.68–1.05), the improvement was not statistically significant at this data cut-off and will be evaluated further with longer follow-up[108]. The ongoing phase I LITESPARK-010 trial (NCT05030506) is also evaluating belzutifan plus lenvatinib (at the same doses), with or without pembrolizumab, specifically in Chinese patients with previously treated advanced-stage ccRCC. Among 24 patients (19 ICI-naive) enrolled in cohort 1 testing the combination without pembrolizumab, the ORR was 50%, median PFS was 13.7 months, median DOR was 17.1 months and median OS was not reached after a median follow-up of 32.1 months[109,110] (Table 1).

In the phase Ib–II KEYMAKER-U03 Substudy 03A, pembrolizumab combined with lenvatinib plus belzutifan was evaluated as first-line therapy for advanced-stage ccRCC, irrespective of *VHL* status. Among 80 efficacy-evaluable patients, this triplet resulted in an ORR of 77.5% with a 12.5% CR rate. The median DOR was 33.4 months, and median PFS was 31.8 months with a 24-month PFS of 67.1%; median OS had not yet been reached after a median 25.7 months of follow-up, with a 24-month OS of 85.5%[111]. Although cross-trial comparisons are inherently limited, the ORR and PFS reported with the triplet are numerically within the upper range of those reported for contemporary first-line ICI–TKI regimens[112–114]. Notably, median DOR and PFS with the triplet were numerically longer compared with the concurrent pembrolizumab–lenvatinib reference arm (25.6 and 26.3 months, respectively) in KEYMAKER-U03, despite a similar ORR (80.6%)[111]. Another substudy of KEYMAKER-U03, 03C (NCT07049926), is evaluating belzutifan with or without the multi-kinase inhibitor zanzalintinib in patients with unresectable locally advanced or metastatic ccRCC that has relapsed during or within 24 months of adjuvant ICI therapy (Supplementary Table 1). A safety lead-in phase will establish the recommended phase II dose of the combination, then 120 patients will be randomly assigned (2:1) to receive belzutifan plus zanzalintinib or belzutifan alone. The primary end points include safety and ORR by blinded independent central review. In parallel, the phase III LITESPARK-033 trial (NCT07227402) is evaluating the combination of belzutifan and zanzalintinib versus cabozantinib in patients with advanced-stage RCC that has recurred during adjuvant anti-PD-(L)1 therapy or within 24 months after its completion (Supplementary Table 1). This randomized study aims to enrol approximately 900 patients and has PFS and OS as co-primary end points.

Other ongoing efforts testing belzutifan triplets include the phase III LITESPARK-012 trial (NCT04736706), which is evaluating the addition of belzutifan or the anti-CTLA4 antibody quavonlimab to pembrolizumab plus lenvatinib as first-line therapy in patients with advanced-stage ccRCC[115] (Supplementary Table 1). This trial has completed accrual and results are eagerly awaited.

As mentioned previously, one of the expansion cohorts of the ongoing phase I–Ib ARC-20 trial is evaluating the safety and efficacy of casdatifan (100 mg) plus cabozantinib (60 mg), both administered orally once daily, in patients with advanced-stage ccRCC previously treated with anti-PD-(L)1 antibody-based therapy (NCT05536141). Preliminary data from 24 efficacy-evaluable patients have been reported, demonstrating a confirmed ORR of 46% (including 1 CR), with only 1 patient (4%) having progressive disease as the best response[116,117]. The combination had a manageable safety profile, with AEs consistent with the known profiles of each agent. Grade ≥3 TRAEs occurred in 48% of patients, with anaemia (24%), hypoxia (7%) and hyponatraemia (7%) being the most common[117]. Notably, no grade 4 or 5 AEs were attributed to casdatifan. Dose reductions owing to AEs were needed in 24% of patients for casdatifan and 38% for cabozantinib[117]. Only two patients discontinued any study drug owing to AEs, and no patient discontinued both agents[116]. To further explore this combination, the phase III PEAK-1 trial (NCT07011719; Supplementary Table 1) is comparing casdatifan (100 mg) plus cabozantinib (60 mg) versus cabozantinib monotherapy in patients with advanced-stage ccRCC that has progressed after anti-PD-(L)1 antibody therapy. This trial is currently open for accrual.

Additionally, the phase Ib–II STELLAR-009 (NCT06191796) trial was initiated to evaluate casdatifan in combination with zanzalintinib, a novel, multi-targeted TKI with activity against VEGFR, MET and the TAM kinases (TYRO3, AXL, MER), as well as the triplet combination of casdatifan, zanzalintinib and nivolumab in patients with advanced-stage solid tumours, including ccRCC. However, this trial has been terminated prior to starting the phase II portion owing to a business decision by the sponsor (Supplementary Table 1).

Immune-checkpoint inhibitors

Preclinical data indicate that combining the HIF2α inhibitor PT2385 with anti-PD-(L)1 or anti-CTLA4 antibodies has additive or even synergistic activity in inhibiting tumour growth in certain syngeneic mouse models, despite the fact that these tumour cell lines did not express HIF2α, suggesting a potential immunosensitizing role for HIF2α inhibition beyond canonical pseudo-hypoxic tumour types[118]. Another study showed that combining casdatifan with the anti-PD-1 antibody zimberelimab produced faster and deeper xenograft tumour regressions, as well as more sustained control when stopping therapy, compared with either agent alone[119]. Furthermore, the non-overlapping immune-related AEs of ICIs and non-immune toxicities of HIF2α inhibitors, such as anaemia and hypoxia, are encouraging for clinical development of this combinatorial approach.

Clinically, this ICI–HIF2α inhibitor approach has been evaluated in an expansion cohort of a phase I trial, in which 50 patients with previously treated advanced-stage ccRCC (one to three prior lines) received PT2385 (800 mg twice daily) in combination with nivolumab (3 mg/kg every 2 weeks). The regimen was well tolerated and antitumour activity was observed (ORR 22%)[120]. Notably, patients with predicted therapeutic levels of PT2385 exposure (plasma trough levels at steady state ≥300 ng/ml; $n = 33$) had longer PFS (10.0 months versus 4.7 months in those with subtherapeutic exposure)[120]. Following the findings of KEYNOTE-564, which demonstrated an OS benefit with adjuvant pembrolizumab in patients with completely resected ccRCC[121], the phase III LITESPARK-022 trial (NCT05239728) evaluated whether adding belzutifan to pembrolizumab improves outcomes in the adjuvant setting after nephrectomy in patients with intermediate–high or high-risk ccRCC as well as in those with metastatic disease and no evidence of disease following nephrectomy and metastasectomy[122]. Among 1,841 enrolled patients, the combination significantly improved disease-free survival (DFS) compared with pembrolizumab alone (HR 0.72, 95% CI 0.59–0.87; $P = 0.0003$)[123]. Median DFS was not reached in either arm at a median follow-up of 28.4 months, although the estimated 24-month DFS was 80.7% with pembrolizumab plus belzutifan versus 73.7% with pembrolizumab plus placebo[123]. OS data were immature at this interim analysis, with a total of 87 OS events[123]. TEAEs of grade ≥3 were more frequent with the belzutifan combination compared to pembrolizumab plus placebo (incidence 52.1% versus 30.2%), most commonly anaemia (12.1% versus 0.4%), increased serum alanine aminotransferase levels (6.4% versus 2.0%) and hypoxia (4.6% versus 0%)[123]. Grade 5 TEAEs occurred in 1.1% and 1.2% of patients in the experimental and control arms, respectively, while treatment-related grade 5 events were reported in 0.3% of patients in each group[123].

Another arm of the KEYMAKER-U03 Substudy 03B evaluated the combination of pembrolizumab and belzutifan (120 mg once daily) in patients with advanced-stage ccRCC that progressed on or after treatment with an anti-PD-(L)1 antibody and VEGFR TKI therapy (median three prior lines)[106]. Among 62 patients, the combination of pembrolizumab and belzutifan was associated with an ORR of 23% (including two CRs)[106]. At a median follow-up of 22.9 months, median PFS was 5.5 months, median DOR was 23.6 months and median OS was 22.9 months[106]. This combination was associated with a 9% rate of grade ≥3 AEs[106]. The phase Ib–III eVOLVE-RCC02 trial (NCT07000149; Supplementary Table 1) is investigating casdatifan combined with volrustomig, a bispecific antibody targeting both PD-1 and CTLA4. In phase Ib of this trial, patients with advanced-stage ccRCC are being randomly assigned to receive one of two dose levels of volrustomig in combination with casdatifan. Dose-finding based on AEs will guide progression to phase III, in which the recommended phase III dose combination will be compared to volrustomig monotherapy and to nivolumab plus ipilimumab in the first-line setting.

Belzutifan is also being investigated in the neoadjuvant setting in BLAZE (NCT07187778; Supplementary Table 1), an ongoing phase II trial evaluating belzutifan, pembrolizumab or their combination prior to nephrectomy in patients with intermediate–high-risk, high-risk or oligometastatic ccRCC. Patients are receiving 12 weeks of systemic therapy prior to surgery. The primary end point is a composite measure of objective radiographic response or pathological response (defined as ≤25% residual viable tumour). The trial plans to enrol 10 patients.

Data published in 2025 indicate that HIF transcription factors, including HIF2α, promote the expression and translation of endogenous retroviral elements (ERVs) in cancer cells, thereby enhancing tumour immunogenicity via viral mimicry and type I interferon signalling[124]. Notably, pharmacological inhibition or genetic depletion of HIF2α led to reduced ERV expression and diminished immune activation[124]. These findings raise the possibility that HIF2α inhibition could attenuate antitumour immunity and thus reduce the efficacy of ICIs; however, HIF2α antagonism might also alleviate hypoxia-driven immunosuppressive mechanisms and improve the tumour immune microenvironment. Therefore, the efficacy of this combination approach will ultimately need to be clarified with the outcomes of the aforementioned ongoing clinical trials (Supplementary Table 1).

CDK4/6 inhibitors

Transition from the G1 to S phase of the cell cycle is regulated by the tumour-suppressor retinoblastoma-associated protein (pRb; also known as RB1), which inhibits cell-cycle progression unless it is

sequentially phosphorylated by cyclin-dependent kinases (CDKs). Loss of chromosome 3p, containing the *VHL* locus, is associated with upregulation of pRb[125]. In a study evaluating 217 RCC samples, including 174 ccRCC samples, pRb expression was observed in ~95% of tumours; however, the potentially tumour-suppressive effects of pRb expression can be overcome by HIF2α, which promotes the expression of cyclin D1 and CDK4 (refs. 126,127). Cyclin D1 activates CDK4 and/or CDK6 (CDK4/6) to phosphorylate pRb, rendering *VHL*-mutant ccRCC cells more highly dependent on CDK4/6 compared with *VHL*-wild-type cells[128]. Accordingly, the antiproliferative effects of the CDK4/6 inhibitors abemaciclib and palbociclib are synergistic with HIF2α inhibition in HIF2α-dependent $VHL^{-/-}$ ccRCC cell lines[128].

A phase Ib trial evaluated abemaciclib monotherapy in heavily pretreated, biomarker-unselected patients with sporadic ccRCC as a prelude to the combination of abemaciclib with belzutifan (NCT04627064). Among 11 patients, abemaciclib monotherapy did not induce any objective responses and was associated with a median PFS of only 1.8 months[129]. Therefore, the combination arm incorporating belzutifan was never initiated. A key limitation of this trial was the absence of selection for *VHL* alterations or other markers of HIF2α dependence, which might have reduced the likelihood of benefit. Nevertheless, LITESPARK-024 (NCT05468697), an open-label, multicentre, randomized phase I–II trial, is evaluating the combination of belzutifan plus palbociclib in patients with advanced-stage sporadic RCC who have received two or more prior lines of therapy (including an anti-PD-(L)1 antibody and a VEGFR TKI in sequence or in combination)[130]. Preliminary results indicate that belzutifan plus palbociclib (75, 100 or 125 mg) has a manageable safety profile but is associated with high rates of grade ≥3 TRAEs (65.0–78.9%), most commonly anaemia (50.0–57.9%), with treatment discontinuation owing to TRAEs occurring in 5.3–26.3% of patients[131]. ORRs were 15.0%, 0% and 21.1% across the 75, 100 and 125 mg palbociclib cohorts, with corresponding median PFS durations of 7.2, 5.4 and 9.1 months[131]. Further development of the regimen seems unlikely considering that the randomized phase portion of this trial will no longer be pursued. Another phase II trial that aimed to evaluate imdatifan as a doublet with palbociclib and as a triplet with palbociclib and the anti-PD-1 antibody sasanlimab has been terminated as a result of sponsor portfolio reprioritization (NCT05935748).

GCN2 activators

Activation of the serine–threonine-protein kinase GCN2 results in phosphorylation of eukaryotic initiation factor 2α (eIF2α), thus markedly repressing general RNA translation (albeit while enhancing the translation of a specific subset of mRNAs) in the setting of an integrated stress response[132]. In preclinical studies, HC-7366, a highly selective and potent activator of GCN2, demonstrated antitumour activity across models of various tumour types, including RCC, and was associated with suppression of both HIF1α and HIF2α signalling[133]. These observations motivated combination studies with belzutifan in models of HIF2-dependent RCC, in which the combination enhanced antitumour efficacy compared with belzutifan alone[133]. Based on these xenograft data, a phase Ib, open-label, multicentre trial (NCT06234605; Supplementary Table 1) has been initiated to evaluate belzutifan in combination with HC-7366 in patients with advanced-stage ccRCC (regardless of *VHL* status), with the primary objective of establishing the safety, tolerability, maximum tolerated dose and recommended phase II dose of this combination[134].

Current clinical challenges with HIF2α inhibitors

Toxicity management

Inhibiting HIF2α leads to dose-dependent suppression of EPO synthesis, resulting in decreased erythropoiesis[83,135]. Consequently, anaemia emerges as the most prevalent on-target AE associated with HIF2α inhibitors[85,90]. Notably, this contrasts with HIF prolyl hydroxylase inhibitors (HIF-PHIs), which stabilize HIF and are approved in many countries to treat anaemia associated with chronic kidney disease, highlighting the double-edged sword of HIF pathway modulation[136]. In a pooled analysis encompassing 576 patients with RCC across four clinical trials testing belzutifan (LITESPARK-001, LITESPARK-004, LITESPARK-005 and LITESPARK-013), anaemia of any grade was reported in 84.2% of patients and was of grade 3–4 in 28.8%[137]. The median time to onset of anaemia was 29 days[137]. Management strategies included the use of ESAs alone in 22.9% of the 485 affected patients, blood transfusions alone in 17.5% and a combination of both in 12.8%[137]. Current FDA prescription information recommend withholding belzutifan if haemoglobin levels fall below 8 g/dl, with blood transfusion as clinically indicated; belzutifan can be resumed at a reduced dose upon recovery or discontinued altogether depending on the clinical scenario[138]. This recommendation was derived from the protocols of pivotal trials of belzutifan; however, supporting clinical data remain lacking. Available long-term safety data demonstrate that dose reductions and supportive treatment might be effective in maintaining tolerability without compromising disease control[139,140]. In a single-centre study involving 25 patients with VHL-associated neoplasms receiving belzutifan, none of those requiring dose reductions ($n = 15$) had treatment failure during a median follow-up of 21.3 months[139]. Similarly, in the 50-month follow-up analysis of the LITESPARK-004 trial in the VHL syndrome population, clinical outcomes were similar between participants who received ESAs and those who did not, suggesting that supportive management does not adversely affect the antitumour activity of belzutifan[140].

Hypoxia is another commonly reported AE of HIF2α inhibitors, although the underlying mechanism remains poorly understood and is probably multifactorial. The PHD2–HIF2 signalling pathway has been implicated in the regulation of the ventilatory response[141]. Preclinical studies in mice have shown that treatment with PT2385 prevents the HIF2α-mediated morphological and proliferative responses of carotid body cells to hypoxia, thereby impairing hypoxia-mediated increases in ventilatory sensitivity[142]. HIF2α is also involved in hypoxic lung vasoconstriction as evidenced in patients with Chuvash disease. This autosomal recessive congenital disease is caused by 598C>T mutations in exon 3 of both *VHL* alleles, which results in a R200W substitution in pVHL that hinders its interaction with HIFα subunits, ultimately causing polycythaemia but notably also pulmonary hypertension[143,144]. In $Vhl^{R/R}$ mouse models of Chuvash disease (homozygous for the R200W mutation), HIF2 activity in the lung was increased, and HIF2 heterozygosity prevented the development of erythrocytosis and pulmonary hypertension[144]. Belzutifan was also effective in mitigating pulmonary fibrosis and hypertension in this model via decreases in endothelin 1 and CXCL12 expression[145]. Therefore, the impairment of hypoxic pulmonary vasoconstriction by HIF2α inhibitors might be involved in worsening V/Q (ventilation/perfusion) mismatch and subsequent hypoxaemia[146]. Consistent with this possibility, a clinical report described belzutifan-associated hypoxaemia that emerged only at high altitude, with an increased shunt fraction in the absence of an anatomical shunt, that resolved at sea level, suggesting a dynamic physiological right-to-left shunt owing to impaired hypoxic adaptation[147]. In the aforementioned pooled analysis of 576 patients treated with

belzutifan[137], hypoxia of any grade was observed in 16.3%, with grade 3 or 4 hypoxia reported in 12.2%. The median time to onset of hypoxia was 31 days, and 70.2% of the 94 affected patients received supplemental oxygen therapy. Notably, substantially higher rates of hypoxia were reported among 34 patients residing at a mean altitude of 1,457 metres above sea level who were treated with belzutifan for metastatic RCC: grade ≥1 hypoxia occurred in 88% of these patients, grade 3 hypoxia in 47%, and 82% ultimately required supplemental oxygen at home[148]. Indeed, management of hypoxia involves monitoring oxygen saturation levels and providing supplemental oxygen as needed. For patients with decreased oxygen saturation during rest (pulse oximeter reading <88% or partial pressure of oxygen in arterial blood (PaO_2) ≤55 mmHg), belzutifan should be interrupted until saturation recovers to >88%, then resumed at a reduced dose[138]. If hypoxaemia occurs during exercise, withholding belzutifan is reasonable; when hypoxaemia resolves, belzutifan can then be restarted at the same or a reduced dose[138]. Permanent discontinuation is considered for life-threatening or recurrent hypoxic events[138]. Clinical trials of belzutifan excluded patients with baseline hypoxaemia (pulse oximeter <92%) or with a chronic or intermittent need for supplemental oxygen[90]. Therefore, future studies are needed to investigate the safety of HIF2α inhibitors in patients with pre-existing pulmonary conditions.

Other frequently reported HIF2α inhibitor-related AEs included fatigue (occurring in 42.7% of patients receiving belzutifan in the pooled analysis), nausea (24.1%) and dyspnoea (21.4%); however, these AEs are less specific and generally low grade (grade 3–4 in only 2.8%, 0.9% and 1.7%, respectively)[137].

Biomarkers of response and resistance

As HIF2α antagonists become increasingly integrated into the treatment landscape of ccRCC, predictive biomarkers of sensitivity or resistance will be crucial for patient selection as well as to better understand the mechanisms of response. Use of HIF2α inhibitors at earlier disease stages, guided by such predictive biomarkers, might also optimize therapeutic sequencing and limit unnecessary toxicity from ineffective combinations.

In a biomarker analysis performed as part of a phase I–II trial testing imdatifan in patients with metastatic ccRCC, low baseline tumour cell expression of the HIF2 transcriptional target GLUT1 on immunohistochemistry was associated with progressive disease ($P = 0.045$) and significantly shorter PFS (56 days versus 505 days; $P = 0.054$), whereas higher GLUT1 expression was correlated with clinical benefit and with *VHL* truncating mutations[149]. By contrast, the expression of cyclin D1, another HIF2 transcriptional target, was not associated with treatment outcomes[149]. Similarly, exploratory biomarker analyses from the phase II LITESPARK-013 trial evaluating two doses of belzutifan in previously treated patients with advanced-stage ccRCC suggest that tumoural HIF2α expression might correlate with clinical outcomes[150]. Among 154 enrolled patients, HIF2α immunohistochemistry data were available for 136 patients, RNA-sequencing data for 109 and whole-exome sequencing data for 113. Higher HIF2α tumour proportion score and H-score on immunohistochemistry were significantly associated with longer PFS ($P = 0.016$ and $P = 0.012$, respectively), although no associations were observed with ORR or OS[150]. RNA-sequencing data showed that higher cancer cell HIF2α transcriptional scores were associated with ORR at the prespecified significance threshold ($P = 0.085$; multiplicity-adjusted $\alpha = 0.10$), whereas HIF2α co-expression scores were not associated with ORR, PFS or OS[150]. Whole-exome sequencing did not identify genomic predictors of response overall; however, the ORR was numerically higher among tumours harbouring *BAP1* mutations compared with *BAP1*-wild-type tumours[150].

In ARC-20, casdatifan produced rapid (as early as 4 h), deep and sustained decreases from baseline serum levels of EPO — another direct transcriptional target of HIF2α[151]. Greater reductions in serum EPO levels as well as higher tumoural EPO expression were strongly correlated with objective responses and a lower likelihood of primary resistance (that is, progressive disease as best response), supporting serum EPO as a pharmacodynamic biomarker of HIF2α inhibition and suggesting that tumoural EPO expression could be enriched for patients more likely to benefit[94,151].

In a retrospective genomic analysis of patients with RCC who received belzutifan, 240 patients were identified and stratified according to time on treatment (using the median of 77 days as a threshold) into responders ($n = 109$) and non-responders ($n = 131$)[152]. Responders had numerical enrichment of alterations in the PI3K–AKT–mTOR pathway, including mutations in *PIK3CA* (12.4% versus 3.3%), *PIK3CB* (6.7% versus 1.9%), *PIK3R1* (4.8% versus 0.8%) and *PTEN* (13.2% versus 9.8%)[152]. Alterations in DNA damage response genes were also more frequent among responders, particularly those in *STAG2* (6.7% versus 0%) and *ATM* (4.8% versus 0.8%)[152]. By contrast, non-responders had numerically higher frequencies of mutations affecting chromatin remodelling genes, including *BAP1* (17.7% versus 12.2%), *ARID1A* (8.3% versus 2.9%), *NF2* (4.0% versus 1.9%) and *KDM5C* (14.6% versus 11.8%), although none of these differences reached statistical significance[152]. Transcriptomic analyses also revealed modest numerical differences in HIF pathway gene expression, including slightly higher expression of *EPAS1* (median $\log_2$[TPM+1] 7.73 versus 7.54) and *CA9* (4.82 versus 4.46) in responders versus non-responders[152]. Furthermore, in a real-world analysis involving 39 patients with available genomic profiling data, *VHL* alterations were identified in 76.9% of cases, and their presence was associated with longer median OS (484 days versus 96 days) and a slightly longer time on treatment (113 days versus 95 days) with belzutifan, although these differences were not statistically significant[153].

Resistance to HIF2α inhibitors has also been linked with differences in HIF2α expression and downstream transcriptional programmes. In mouse xenografts stratified by sensitivity to PT2399, sensitive tumours had strong HIF2α nuclear staining and high levels of HIF2α protein expression, whereas resistant tumours had markedly lower HIF2α levels across all assays; resistant tumours also had substantially lower *EPAS1* mRNA levels, indicating loss of HIF2α dependence[79]. Moreover, integrative profiling identified differentially expressed genes that discriminate sensitive from resistant phenotypes, including *GLI1*, *PTHLH*, *EZH2*, *HIF1A* and *MCAM*[79], reflecting a transcriptional landscape that might bypass HIF2 signalling. The expression of HIF1α on immunohistochemistry was also notably increased in resistant tumours compared to sensitive ones, suggesting a potential compensatory HIF1α-driven switch in response to HIF2 inhibition[79].

Moreover, mutations in components of the HIF2 heterodimer might confer resistance by impairing inhibitor binding or by stabilizing heterodimer interfaces. For example, a heterozygous *EPAS1* c.968G>A mutation produces a G323E substitution within the PAS-B binding pocket of HIF2α, which probably creates steric hindrance that prevents the binding of small-molecule HIF2α inhibitors[74,79,154]. Additionally, a mutation in *ARNT* (c.1338C>A) results in an F446L substitution within the PAS-B domain of HIF1β[79], which enhances HIF2α–HIF1β dimer stability[155]. Unlike the G323E mutation, the F446L substitution was hypothesized to confer only partial resistance to HIF2α inhibition, if any, as PT2385 retained concentration-dependent disruption of

the HIF2α–HIF1β heterodimer (K_i of 127 nM for F446L-mutant versus 148 nM for wild-type HIF1β), consistent with maintained binding of PT2385 within the target pocket in HIF2α[155]. Non-genetic mechanisms have also been described, including phosphorylation of HIF2α at T324, which impairs belzutifan binding through allosteric effects in vitro, although its clinical relevance remains uncertain[156].

Beyond direct HIF2 pathway alterations, mutations in *TP53* were initially linked with resistance to HIF2α inhibitors in RCC cell lines[80] and in one patient with ccRCC who had an acquired *TP53* mutation identified following disease progression on PT2385 (ref. 157). However, subsequent studies demonstrated that *TP53* mutational status does not alter sensitivity to HIF2α antagonists[158].

Several strategies to overcome resistance to HIF2α inhibitors are under investigation. Targeting HSP70, a molecular chaperone that stabilizes HIF2α, can restore sensitivity to belzutifan in cell lines[159]. Inhibiting the peptidyl-prolyl *cis-trans* isomerase FKBP10, which is upregulated upon HIF2α inhibition and promotes metabolic remodelling by enhancing lactate dehydrogenase A phosphorylation, also enhances the therapeutic efficacy of PT2385 in ccRCC models[160]. Future research should focus on clarifying the mechanisms of both primary and acquired resistance to HIF2α inhibitors and developing rational combination strategies or novel inhibitors to overcome this challenge.

Expanding the indications for HIF2α inhibitors

While HIF2α is well established as a key driver in ccRCC and VHL-associated tumours, emerging evidence indicates that this transcription factor can also contribute to tumour progression in other malignancies, including breast, colorectal, liver and prostate cancers[161–164]. These findings suggest a role for HIF2α inhibitors beyond kidney cancer.

Pheochromocytomas and paragangliomas

Belzutifan has been used for the treatment of patients with an early postzygotic somatic mutation in *EPAS1*, specifically the gain-of-function p.A530V pathogenic variant that impairs the regulatory P531 hydroxylation of HIF2α. These patients often have polycythaemia, multiple abdominal PPGL tumours and paroxysmal hypertension, collectively known as Pacak–Zhuang syndrome. In an initial case report describing a patient receiving belzutifan for this syndrome, rapid reductions in plasma catecholamine levels were observed within 9 days, with normetanephrine concentrations decreasing from 16.2 to 5.7 nmol/l, and polycythaemia resolution with haemoglobin normalization by day 17 (ref. 165). Tumour shrinkage was also evident by day 17, with sustained response over 24 months of treatment[165]. Additional case reports provide further evidence of these rapid cardiovascular effects of belzutifan, resulting in improved acute management of catecholamine-driven crises in patients with Pacak–Zhuang syndrome, with plasma norepinephrine levels normalizing within 3–6 h and blood pressure stabilizing correspondingly, a response time far superior to existing therapies such as metyrosine[166].

Beyond Pacak–Zhuang syndrome, cohort A1 of LITESPARK-015 included 72 patients with progressive locally advanced or metastatic PPGL not amenable to curative-intent treatment[167]. At a median follow-up of 30.2 months, belzutifan conferred a confirmed ORR of 26% and a disease control rate of 85% in this cohort, with durable benefit reflected by a median DOR of 20.4 months and median PFS of 22.3 months[167]. Belzutifan also demonstrated clinically meaningful endocrine control, with 32% of patients (19 of 60) receiving baseline antihypertensive therapy having a sustained ≥50% reduction in at least one antihypertensive medication for ≥6 months. The encouraging results from LITESPARK-015 led to the FDA approval of belzutifan in May 2025 for patients ≥12 years of age with unresectable advanced-stage PPGL, providing the first approved oral therapy for this disease[4].

Notably, *EPAS1*-driven PPGL seems to be particularly responsive to belzutifan. Real-world data from five patients with PPGL carrying somatic *EPAS1*-activating variants demonstrated an ORR of 80% (four PRs, with stable disease in the remaining patient), with an average tumour diameter reduction of 36.8%[168]. Importantly, polycythaemia resolved in all treated patients with prior erythrocytosis, eliminating the need for therapeutic phlebotomy, with plasma EPO levels declining by 97% and haemoglobin levels by 13%[168]. Furthermore, catecholamine normalization enabled discontinuation or reduction of antihypertensive medications in two patients[168].

Exploratory trials in other solid tumour types

The ongoing phase II LITESPARK-016 trial (NCT04976634) is evaluating the combination of pembrolizumab plus lenvatinib and belzutifan across a range of solid tumour types, including hepatocellular carcinoma, colorectal cancer, pancreatic ductal adenocarcinoma, biliary tract cancer, endometrial cancer and oesophageal squamous cell carcinoma[169]. In addition, several other ongoing trials are testing HIF2α inhibitors in specific cancers beyond those classically associated with genetic alterations directly affecting the VHL–HIF signalling pathway (Supplementary Table 1).

Breast cancer. Preclinical evidence supports a role of HIF2α antagonism in suppressing the growth of tamoxifen-resistant breast cancer cell lines, particularly when combined with the selective oestrogen receptor (ER) degrader fulvestrant[170,171]. Building upon these findings, the open-label, multicentre, randomized, phase II LITESPARK-029 trial (NCT06428396; Supplementary Table 1) is comparing belzutifan (120 mg daily) plus fulvestrant versus everolimus plus either fulvestrant or exemestane in patients with advanced-stage ER^+, $HER2^-$ breast cancer[172].

Hepatocellular carcinoma. MORPHEUS-Liver (NCT04524871; Supplementary Table 1) is a randomized phase Ib–II umbrella trial evaluating various immunotherapy-based combinations in patients with advanced-stage hepatocellular carcinoma. In one cohort, imdatifan in combination with the anti-PD-L1 antibody atezolizumab plus the anti-VEGFA antibody bevacizumab will be compared with atezolizumab plus bevacizumab alone in the first-line setting.

Glioblastoma. HIF2α is frequently overexpressed in glioblastoma, particularly in high-grade tumours and glioma stem cells within hypoxic perivascular and perinecrotic regions, and correlates with poor patient outcomes[173,174]. HIF2α inhibition has been shown to modulate the immunosuppressive properties of the tumour microenvironment and decrease tumour volume in mouse models of glioblastoma[175]. The efficacy of PT2385 monotherapy in patients with recurrence of this disease after first-line chemoradiotherapy has been tested in a two-stage, single-arm phase II trial[176]. No objective radiographic responses were observed in the 24 patients enrolled, although the 3 patients with the highest levels of systemic exposure to PT2385 (day 15 C_{min} >1,000 ng/ml) had a significantly longer median PFS than those with lower exposure (6.7 months versus 1.8 months; $P = 0.009$)[176]. Similarly, no objective responses were observed among 25 patients included in the glioblastoma expansion cohort of the first-in-human phase I–II LITESPARK-001 trial testing belzutifan as a single agent across advanced-stage solid tumour types[177].

Prostate cancer. Evidence suggests that HIF2α expression correlates with a higher Gleason score in patients with prostate cancer[164]. Furthermore, chronic hypoxia has been shown to promote androgen insensitivity and the development of a castration-resistant state in prostate cancer cells[178]. These findings highlight the role of HIF2α in driving aggressive disease biology and mechanisms of resistance, providing a rationale for targeting HIF2α in clinical trials. The phase Ib–II KEYNOTE-365 trial (NCT02861573; Supplementary Table 1) is evaluating the combination of pembrolizumab with other therapies in patients with metastatic castration-resistant prostate cancer after disease progression on prior chemotherapy. Belzutifan is being tested in cohort J of this trial: in the phase Ib lead-in, 20 patients will receive belzutifan as monotherapy; if responses are observed, the phase II expansion will randomly assign up to 180 more patients (1:1) to belzutifan plus pembrolizumab versus belzutifan alone[179].

Pancreatic adenocarcinoma. Pancreatic adenocarcinoma (PDAC) is characterized by an immunosuppressive desmoplastic stroma that impairs drug delivery, rendering the disease highly resistant to most systemic treatments and particularly to immunotherapies[180,181]. Deletion of HIF2α specifically in cancer-associated fibroblasts has been shown to inhibit tumour growth and improve survival in mouse models of PDAC, in association with reduced infiltration by immunosuppressive cells such as M2-like macrophages and regulatory T cells[182]. Repurposing HIF2α inhibitors in patients with PDAC could therefore prove beneficial in terms of enhancing tumour responsiveness to ICI-based regimens. LITESPARK-016 (NCT04976634) is a phase II trial evaluating belzutifan in combination with pembrolizumab and lenvatinib across various solid tumour types, including a dedicated cohort of patients with advanced-stage PDAC.

Sporadic pancreatic neuroendocrine tumours. Whereas VHL-associated pNETs are highly sensitive to HIF2α inhibition[86,88], belzutifan has shown substantially more limited activity in patients with sporadic pNETs. Cohort A2 of LITESPARK-015 enrolled 70 patients with advanced-stage, well-differentiated (grade 1–2) pNETs that had progressed on prior targeted therapy[183]. Treatment of these patients with belzutifan (120 mg daily) resulted in an ORR of only 10%, a disease control rate of 66% and a median PFS of 3.9 months[183]. However, the median DOR and median OS were not reached at a median follow-up duration of ~31 months[183].

Clear-cell gynaecological malignancies. Clear-cell gynaecological malignancies, including ovarian and endometrial clear-cell carcinomas, are uncommon but clinically aggressive tumour subtypes that often have limited responses to conventional chemotherapy. An ongoing phase II, open-label trial (NCT06677190; Supplementary Table 1) is evaluating belzutifan in patients with recurrent or persistent clear-cell gynaecological malignancies, including ovarian, endometrial and cervical cancers with a clear-cell histology. This trial is expected to enrol 30 patients.

Paraneoplastic syndromes

ccRCC frequently causes paraneoplastic syndromes, such as hypercalcaemia and cachexia, which are associated with aggressive disease and poor prognosis. The molecular mechanisms underlying these systemic manifestations have historically not been completely understood[184,185]. *PTHLH*, the gene encoding parathyroid hormone-related protein (PTHrP), is known to be a direct transcriptional target of HIF2 (ref. 186). More recent data indicate that HIF2 binding to HREs in the *PTHLH* promoter drives increased PTHrP expression, which mediates humoral hypercalcaemia of malignancy and contributes to systemic metabolic alterations[187]. In RCC xenograft models characterized by hypercalcaemia, pharmacological inhibition of HIF2α using PT2399 reduced HIF2 occupancy at the *PTHLH* promoter, decreased circulating PTHrP levels, and normalized serum calcium levels[187]. Complementary studies in orthotopic tumour models demonstrated that increased *PTHLH* expression is sufficient to induce both hypercalcaemia and cancer-associated cachexia[187]. Importantly, early clinical observations indicate that HIF2α inhibition can rapidly reverse these systemic manifestations: treatment with HIF2α inhibitors, such as belzutifan or NKT2152, has been associated with rapid correction of hypercalcaemia and improvement in cachexia in patients with ccRCC, in some cases even in the absence of measurable tumour regression[188].

Conclusions

The development of HIF2α inhibitors represents a paradigm of successful bench-to-bedside translation. Specifically, belzutifan has emerged as a treatment option across several tumour types driven by aberrant HIF2 signalling, including ccRCC, VHL-associated neoplasms and PPGL, and serves as proof of concept for expanding this therapeutic strategy to more diverse clinical contexts. The development of novel HIF2α inhibitors, RNAi-based approaches, and combinatorial regimens with ICIs, VEGFR TKIs or modulators of other cellular pathways, such as ferroptosis and integrated stress responses, underscores the likelihood that HIF2 pathway modulation will provide a durable therapeutic platform rather than a niche single-agent approach. Future priorities include the development of predictive biomarkers to quantify HIF2 dependence and anticipate treatment resistance, the design of mechanistically rational, biomarker-integrated trials, and the optimization of strategies to mitigate and manage on-target toxicities of HIF2α inhibitors such as anaemia and hypoxia. Beyond kidney cancer, early signals of activity in a range of hypoxia-adapted malignancies support the continued evaluation of HIF2α antagonists in a broader range of oncological indications. As clinical applications expand, correlative translational efforts will be essential to refine patient selection, optimize the sequencing of HIF2α antagonists with existing standard-of-care treatments, and better define the disease contexts in which HIF2α inhibition confers additive or synergistic benefit.

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E.S. and M.M. researched data for the manuscript and wrote the manuscript. All authors made substantial contributions to discussions of content and reviewed and/or edited before submission.

Competing interests

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